📝 Chapter 3: Hedging Strategies Using Futures (Part 1)

1. Introduction to Hedging with Futures

A. The Goal of Hedging

- **Objective:** To use futures markets to **reduce a specific risk** faced by an individual or company. This risk could be related to commodity prices (oil, copper), interest rates, or foreign exchange rates.
- **Ideal Outcome (Perfect Hedge):** A perfect hedge is one that completely **eliminates the risk** and locks in a known price or cost.
- **Reality:** Perfect hedges are **rare**. Our goal is to construct hedges that perform as close to perfectly as possible.

B. "Hedge-and-Forget" Strategy (Static Hedging)

For now, we are focusing on the simplest hedging strategy:

- **Action:** Take a futures position **at the start** of the hedge's life.
- **Duration:** Hold that position **unchanged**.
- **Closure:** Close out the position **at the end** of the hedge's life.
  (We will cover dynamic, constantly adjusted hedges later in Chapter 19.)

C. Basic Hedging Principle

The core principle of hedging is to take a position that **neutralizes** your existing risk (exposure).

- **Rule:** The gain/loss on your futures position must be designed to **offset** the loss/gain on the underlying asset you are trying to protect.

Existing Exposure (Company Business)	Hedging Futures Position	Outcome
Gain if Price goes ↑	Position designed to **Loss** if Price goes ↑	**Price is locked in.**
Loss if Price goes ↓	Position designed to **Gain** if Price goes ↓	**Price is locked in.**

2. Short Hedges (Section 3.1)

A **short hedge** involves taking a **short position (selling) futures contracts**.

A. When is a Short Hedge Appropriate?

A short hedge is used when you are concerned about a **fall in the price** of an asset.

1. **Current Ownership:** You **own the asset** now and plan to sell it sometime in the future.
   - *Example:* A farmer owns soybeans and plans to sell them at harvest in three months. The farmer is worried the price might fall.
2. **Future Ownership:** You **don't own the asset yet**, but you know you will receive it (or be exposed to it) in the future.
   - *Example:* A U.S. exporter is expecting to receive a payment of €5 **million** in two months. If the euro drops against the dollar, the exporter will receive fewer dollars.

B. Example: Oil Producer Hedging Future Sales (Current Ownership)

Imagine an oil producer who agrees to sell **1 million barrels** in 3 months. The price will be the market price on the delivery date.

- **The Risk:** The producer has a **long exposure** to oil (they own it and gain if the price rises, lose if it falls). They want to lock in the price now.
- **Action:** Short the appropriate futures contracts.
  - *Given:* Spot Price = $80/barrel, August Futures Price = $79/barrel. Contract size = 1,000 barrels.
  - *Calculation:* $1,000,000 \text{ barrels}/1,000 \text{ barrels/contract} = \mathbf{1,000} \text{ futures contracts.}$
  - *Trade:* **Short 1,000 August Crude Oil Futures at $79/barrel.**

Outcome on August 15	Sales Contract Outcome	Futures Position Outcome	Net Result (Approximate)
Price Drops to $75	Realize $75 million for the oil (Loss of $4M vs. initial $79M expectation).	**Gain** of $(\$79 - \$75) \times 1 \text{ million barrels} = \mathbf{\$4} \text{ million}$.	$\$75M + \$4M = \mathbf{\$79} \text{ million.}$ (Price locked in at $79/barrel).
Price Rises to $85	Realize $85 million for the oil (Gain of $6M vs. initial $79M expectation).	**Loss** of $(\$85 - \$79) \times 1 \text{ million barrels} = \mathbf{\$6} \text{ million}$.	$\$85M - \$6M = \mathbf{\$79} \text{ million.}$ (Price locked in at $79/barrel).

- **Conclusion:** In both scenarios, the **short hedge successfully locked in a price close to the August futures price ($79 per barrel)**, eliminating the price risk.

3. Long Hedges (Section 3.1)

A **long hedge** involves taking a **long position (buying) futures contracts**.

A. When is a Long Hedge Appropriate?

A long hedge is used when you are concerned about a **rise in the price** of an asset you will need to buy in the future.

- **Scenario:** A company knows it will have to **purchase an asset** at some time in the future and wants to lock in the cost now.
  - *Example:* A candy manufacturer knows it will need 50,000 pounds of sugar in 4 months. They are worried the sugar price might rise.

B. Example: Copper Fabricator Hedging Future Purchases

Imagine a copper fabricator that needs **100,000 pounds of copper** in 4 months (May 15th) to fulfill a sales contract.

- **The Risk:** The fabricator has a **short exposure** to copper (they must buy it and lose if the price rises). They want to fix the cost.
- **Action:** Long the appropriate futures contracts.
  - *Given:* Spot Price = 340 cents/lb, May Futures Price = 320 cents/lb. Contract size = 25,000 lbs.
  - *Calculation:* $100,000 \text{ lbs}/25,000 \text{ lbs/contract} = \mathbf{4} \text{ futures contracts.}$
  - *Trade:* **Long 4 May Copper Futures at 320 cents/lb ($3.20/lb).**

Outcome on May 15	Spot Purchase Cost	Futures Position Outcome	Net Cost (Approximate)
Price Rises to 325 cents/lb	Purchase costs $100,000 \times \$3.25 =$ $\mathbf{\$325,000}$.	**Gain** of $100,000 \times$ $(\$3.25 - \$3.20) =$ $\mathbf{\$5,000}$.	$\$325,000 - \$5,000 =$ $\mathbf{\$320,000}$. (Cost locked in at 320 cents/lb).
Price Drops to 305 cents/lb	Purchase costs $100,000 \times \$3.05 =$ $\mathbf{\$305,000}$.	**Loss** of $100,000 \times$ $(\$3.20 - \$3.05) =$ $\mathbf{\$15,000}$.	$\$305,000 + \$15,000 =$ $\mathbf{\$320,000}$. (Cost locked in at 320 cents/lb).

- **Conclusion:** In all cases, the **long hedge successfully locked in a net cost close to the May futures price (320 cents/lb)**.

C. Futures vs. Spot Purchase

- In the long hedge example, using futures is clearly superior to buying the copper in the spot market today (at 340 cents/lb) because buying spot would require:
    i. Paying a higher price (340 cents vs. 320 cents).
    ii. Incurring **interest costs** on the funds used to buy the copper today.
    iii. Incurring **storage costs** until the copper is needed in May.

4. Considerations in Basic Hedging

A. Delivery

- **General Practice:** Even when a hedger holds the futures contract until the delivery month, they usually **close out the futures position** rather than making or taking delivery.
- **Reason:** Making or taking delivery can be **costly and inconvenient** (e.g., paying for transport, inspection, warehousing, or dealing with the administrative details).

B. The Role of Daily Settlement

- The basic examples above assume a forward contract (where the entire gain/loss is settled at the end).
- **In Reality (Futures):** Daily settlement means the gain or loss is realized **day by day** throughout the life of the hedge.
- **Effect:** This daily cash flow slightly alters the performance of the hedge due to the time value of money, requiring a minor adjustment known as **"tailing"** (discussed later).